



Rossmoyne Senior High School

Semester One Examination, 2020

Question/Answer booklet

MATHEMATICS APPLICATIONS UNIT 1

Section One:
Calculator-free

SOLUTIONS

WA student number: In figures

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Circle your teacher's name

Buckland

Fletcher

Leonard

Pisano

Smith

Younger

Time allowed for this section

Reading time before commencing work:

five minutes

Working time:

fifty minutes

Number of additional
answer booklets used
(if applicable):

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Materials required/recommended for this section

To be provided by the supervisor

This Question/Answer booklet

Formula sheet

To be provided by the candidate

Standard items: pens (blue/black preferred), pencils (including coloured), sharpener, correction fluid/tape, eraser, ruler, highlighters

Special items: nil

Important note to candidates

No other items may be taken into the examination room. It is **your** responsibility to ensure that you do not have any unauthorised material. If you have any unauthorised material with you, hand it to the supervisor **before** reading any further.

Structure of this paper

Section	Number of questions available	Number of questions to be answered	Working time (minutes)	Marks available	Percentage of examination
Section One: Calculator-free	8	8	50	52	35
Section Two: Calculator-assumed	13	13	100	98	65
Total					100

Instructions to candidates

1. The rules for the conduct of examinations are detailed in the school handbook. Sitting this examination implies that you agree to abide by these rules.
2. Write your answers in this Question/Answer booklet preferably using a blue/black pen. Do not use erasable or gel pens.
3. You must be careful to confine your answers to the specific question asked and to follow any instructions that are specific to a particular question.
4. Show all your working clearly. Your working should be in sufficient detail to allow your answers to be checked readily and for marks to be awarded for reasoning. Incorrect answers given without supporting reasoning cannot be allocated any marks. For any question or part question worth more than two marks, valid working or justification is required to receive full marks. If you repeat any question, ensure that you cancel the answer you do not wish to have marked.
5. It is recommended that you do not use pencil, except in diagrams.
6. Supplementary pages for planning/continuing your answers to questions are provided at the end of this Question/Answer booklet. If you use these pages to continue an answer, indicate at the original answer where the answer is continued, i.e. give the page number.
7. The Formula sheet is not to be handed in with your Question/Answer booklet.

Section One: Calculator-free

35% (52 Marks)

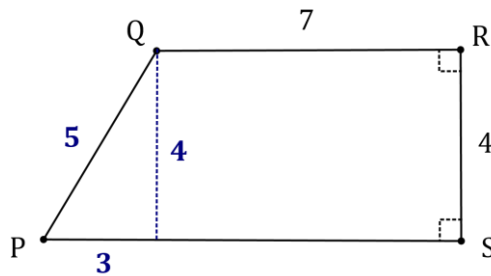
This section has **eight** questions. Answer **all** questions. Write your answers in the spaces provided.

Working time: 50 minutes.

Question 1

(5 marks)

In order to fit a building on a small plot of land, the plan of its concrete slab $PQRS$ is trapezoidal with right-angled corners at R and S . In the diagram shown PS is 3 m longer than QR , QR is 7 m and RS is 4 m.



- (a) Determine the area of the slab $PQRS$.

(2 marks)

Solution
Rectangle: $A = 4 \times 7 = 28$ Triangle: $A = \frac{1}{2} \times 3 \times 4 = 6$ Total area: $A = 28 + 6 = 34 \text{ m}^2$
Specific behaviours
✓ indicates suitable method ✓ correct area

- (b) Formwork is used to surround the edge of the entire slab when concrete is poured and costs \$20 per metre. Determine the cost of the formwork for the slab $PQRS$.

(3 marks)

Solution
$PQ = \sqrt{3^2 + 4^2}$ $= \sqrt{25} = 5$ $P = 5 + 7 + 4 + 10 = 26 \text{ m}$ $\text{Cost} = 26 \times 20 = \520
Specific behaviours
✓ length PQ ✓ total perimeter ✓ correct cost f/t from student's P

Question 2**(5 marks)**

Evaluate the following:

(a) $4 + 5 \times 10.$

(1 mark)

Solution
$4 + 50 = 54$
Specific behaviours
✓ correct value

(b) $12 \div 6 - 5 \times \frac{1}{2}.$

(2 marks)

Solution
$\frac{12}{6} - \frac{5}{2} = 2 - 2.5 = -0.5 \text{ or } -\frac{1}{2}$
Specific behaviours
✓ simplifies quotient and product
✓ correct value

(c) $\frac{\sqrt{3x+4}}{(x-2)^2}$ when $x = 7.$

(2 marks)

Solution
$\frac{\sqrt{3 \times 7 + 4}}{(7 - 2)^2} = \frac{\sqrt{25}}{5^2}$ $= \frac{5}{25}$ $= \frac{1}{5} \text{ or } 0.2$
Specific behaviours
✓ correctly evaluates numerator or denominator
✓ correct, simplified value

Question 3

(7 marks)

When investigating the costs associated with running a 125cc motorbike, a student estimated the following figures for their budget.

Expense	Frequency	
	Monthly (\$)	Half-yearly (\$)
Servicing		90
Registration		60
Insurance		48
Loan repayment	124	
Fuel	85	
Wax and polish	44	

- (a) Briefly explain what the term 'fixed spending' means in relation to a budget and give an example that could apply to the running costs of a motorbike. (2 marks)

Solution
Fixed spending is spending on essential items or needs. The amounts are usually the same each month.
Examples: Spending on loan repayments, insurance, registration, etc.
Specific behaviours
✓ reasonable explanation ✓ satisfactory example

- (b) Briefly explain what the term 'discretionary spending' means in relation to a budget and give an example that could apply to the running costs of a motorbike. (2 marks)

Solution
Discretionary spending is spending on non-essential items or services (spending on "wants" rather than "needs").
Examples: Spending on wax and polish, buying accessories, etc.
Specific behaviours
✓ reasonable explanation ✓ satisfactory example

- (c) Use the figures in the table to determine if a monthly budget of \$350 is enough to cover the running costs of the motorbike. (3 marks)

Solution
$90 \div 6 = \$15$, $60 \div 6 = \$10$, $48 \div 6 = \$8$.
$15 + 10 + 8 + 124 + 85 + 44 = \$286/\text{month}$.
Hence budget of \$350 is more than enough - monthly expenses are much less.
Specific behaviours
✓ converts half-yearly costs to monthly ✓ correct monthly total ✓ states appropriate conclusion

Question 4

(6 marks)

A spreadsheet contains the following values:

	A	B	C	D	E
1	6.5	7.75	6.25	10.25	9.25
2					
3	4.2	4.5	6.2	8.4	9.6
4					

The following formulas are to be used to calculate the value in cell A2 and the value in cell A4:

$$A2 = (A1 + B1 + C1 + D1 + E1) \div 5$$

$$A4 = (0.5 \times A3 + B3 + C3 + D3 + 0.5 \times E3) \div 4$$

(a) Determine the value in cell A2.

(2 marks)

Solution
$A2 = (6.5 + 7.75 + 6.25 + 10.25 + 9.25) \div 5$ $= 40 \div 5$ $= 8$
Specific behaviours
✓ evidence of correct substitution/sum ✓ correct cell value

(b) Determine the value in cell A4.

(2 marks)

Solution
$A4 = (0.5 \times 4.2 + 4.5 + 6.2 + 8.4 + 0.5 \times 9.6) \div 4$ $= (2.1 + 4.5 + 6.2 + 8.4 + 4.8) \div 4$ $= 26 \div 4$ $= 6.5$
Specific behaviours
✓ evidence of correct substitution ✓ correct cell value

(c) The value in cell C1 increased by 15. Determine the change in value of cell A2. (2 marks)

Solution
$A2 = (40 + 15) \div 5$ $= 55 \div 5$ $= 11$ Change in cell value is $11 - 8 = 3$
Specific behaviours
✓ new value of cell A2 ✓ states change in value

Question 5

(7 marks)

When designing an electrical circuit, an engineer uses the table below to determine F , the latency factor. R is the number of resistors and C is the number of capacitors in the circuit.

F		R				
		3	4	5	6	7
C	4	0.33	A	0.49	0.60	0.73
	5	0.39	0.46	0.55	0.66	0.79
	6	0.45	0.52	0.61	0.72	0.85
	7	0.51	0.58	0.67	0.78	0.91
	8	0.57	0.64	0.73	0.84	B

(a) Determine

(i) F when a circuit has 5 resistors and 7 capacitors.

(1 mark)

Solution
$F = 0.67$
Specific behaviours
✓ correct value

(ii) R when a circuit with 4 capacitors has a latency factor of 0.73.

(1 mark)

Solution
$R = 7$
Specific behaviours
✓ correct value

The formula used to calculate the latency factor is $F = \frac{R^2 + 6C}{100}$.

(b) Determine the value of A and the value of B shown in the table.

(2 marks)

Solution
$F_A = (4^2 + 6 \times 4) \div 100 = 0.40$
$F_B = (7^2 + 6 \times 8) \div 100 = 0.97$
Specific behaviours
✓ value of A
✓ value of B

(c) Circuit P has 6 resistors and 11 capacitors and circuit Q has 8 resistors and 8 capacitors. Determine, with justification, which circuit has the highest latency factor.

(3 marks)

Solution
$F_P = (6^2 + 6 \times 11) \div 100 = 1.02$
$F_Q = (8^2 + 6 \times 8) \div 100 = 1.12$
Hence, circuit Q has highest latency factor as 1.12 is greater than 1.02.
Specific behaviours
✓ value of F for circuit P
✓ value of F for circuit Q
✓ correct comment that compares values
For 3 rd mark f/t from above.

Question 6**(7 marks)**

An investor owned 700 shares in two companies, as shown in the table below.

Company	Number of shares	Current market value of share	Dividend per share	Dividend per share
HHA	500	\$6.00	—	3%
MMJ	200	\$2.50	60c	—

- (a) Determine the total market value of these shares.

(2 marks)

Solution
$500 \times 6 = 3000$, $200 \times 2.5 = 500$ Total value: $3000 + 500 = \$3\,500$
Specific behaviours
✓ values for each company ✓ states total value No f/t

- (b) Both companies are about to pay dividends to shareholders. Determine the total dividend that the investor will receive.

(3 marks)

Solution
HHA: $3000 \times 0.03 = 90$
MMJ: $200 \times 0.60 = 120$
Total dividend: $90 + 120 = \$210$
Specific behaviours
✓ first dividend ✓ second dividend ✓ states total dividend No f/t

- (c) Calculate the percent dividend per share (the dividend yield) for MMJ shares.

(2 marks)

Solution
$60c / 250c \times 100\% = 24\%$
Specific behaviours
✓ indicates correct calculation ✓ correct percentage

Question 7

(8 marks)

Consider the matrices $A = \begin{bmatrix} 5 & -2 \end{bmatrix}$, $B = \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix}$ and $C = \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix}$.

(a) Calculate

(i) $B + 4C$.

(1 mark)

Solution
$\begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} + \begin{bmatrix} -12 & -4 \\ 8 & 0 \end{bmatrix} = \begin{bmatrix} -14 & -3 \\ 11 & 4 \end{bmatrix}$
Specific behaviours
✓ correct result

(ii) AC .

(2 marks)

Solution
$\begin{bmatrix} 5 & -2 \end{bmatrix} \times \begin{bmatrix} -3 & -1 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} -19 & -5 \end{bmatrix}$
Specific behaviours
✓ product is correct size ✓ correct product

(iii) B^2 .

(2 marks)

Solution
$\begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} \times \begin{bmatrix} -2 & 1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 7 & 2 \\ 6 & 19 \end{bmatrix}$
Specific behaviours
✓ at least two correct elements in product ✓ correct product

(b) Given that $3B - C + xI = \begin{bmatrix} h & 4 \\ 7 & 20 \end{bmatrix}$, where I is the identity matrix, determine the value of the constant x and the value of the constant h .

(3 marks)

Solution
$3B - C = \begin{bmatrix} -3 & 4 \\ 7 & 12 \end{bmatrix}$
$\begin{bmatrix} -3 & 4 \\ 7 & 12 \end{bmatrix} + \begin{bmatrix} x & 0 \\ 0 & x \end{bmatrix} = \begin{bmatrix} h & 4 \\ 7 & 20 \end{bmatrix}$
$12 + x = 20 \Rightarrow x = 8$
$-3 + 8 = h \Rightarrow h = 5$
Specific behaviours
✓ Demonstrates the xI matrix ✓ value of x ✓ value of h f/t from student's x

Question 8**(7 marks)**

A young person qualifies for a government allowance of \$450 per fortnight, provided their fortnightly income is no more than \$430.

In any fortnight that a person earns more than \$430, the allowance will be reduced by 50 cents in the dollar for earnings over \$430 and up to \$530, and reduced by 60 cents in the dollar for earnings over \$530.

A person who qualifies for the allowance earns \$30 per hour working in an office. Each week they work 5 hours on Tuesday and 4 hours on Wednesday.

- (a) Calculate their fortnightly earnings.

(2 marks)

Solution
Total hours: $(5 + 4) \times 2 = 18$
Earnings per fortnight: $18 \times 30 = \$540$
Specific behaviours
✓ indicates total hours worked in fortnight
✓ correct earnings

- (b) Determine their fortnightly government allowance.

(3 marks)

Solution
First band deduction: $530 - 430 = 100$, $100 \times 0.5 = \$50$
Second band deduction: $540 - 530 = 10$, $10 \times 0.6 = \$6$
Allowance remaining: $450 - 50 - 6 = \$394$
Specific behaviours
✓ deduction for first band
✓ deduction for second band
✓ correct allowance

- (c) Determine the reduction in the fortnightly government allowance calculated in part (b) if they worked an extra 5 hours every Monday.

(2 marks)

Solution
Extra earnings: $5 \times 2 \times 30 = \$300$
Reduction will be: $300 \times 0.6 = \$180$
Specific behaviours
✓ indicates extra earnings
✓ correct reduction

